

CHAPTER 5

THERMOCHEMISTRY

5.1 The Nature of Energy

5.2 The First Law of Thermodynamics

5.3 Enthalpy

5.4 Enthalpies of Reaction

5.5 Calorimetry

5.6 Hess's Law

5.7 Enthalpies of Formation

Thermochemistry

➤ **Thermochemistry:** the study of relationships between chemical reactions and energy changes that involve heat.

➤ **Thermodynamics:** the study of energy and its transformations.

5.1 The Nature of Energy

➤ **Energy (E):** is commonly defined as the capacity to do **WORK (w)** or transfer **HEAT (q)**.

➤ **HEAT** is the energy used to cause the temperature of an object to increase.

➤ **WORK (w)** is the energy used to cause an object to move against a force.

Kinetic Energy

- Kinetic energy (E_k): is the energy of motion.

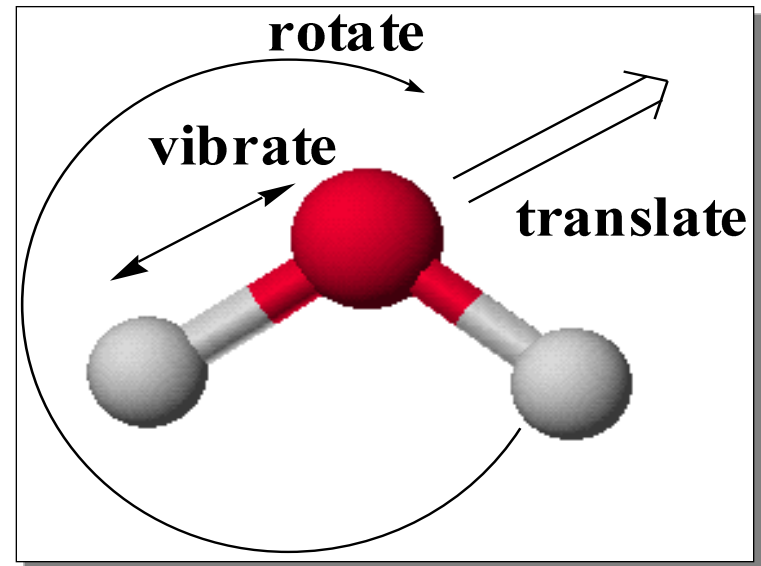
$(\text{Kg} \cdot \text{m}^2 \cdot \text{s}^{-2})$ ← $E_k = \frac{1}{2} \cdot m \cdot v^2$

SI Unit: Joule

Mass (Kg)

Speed ($\text{m} \cdot \text{s}^{-1}$)

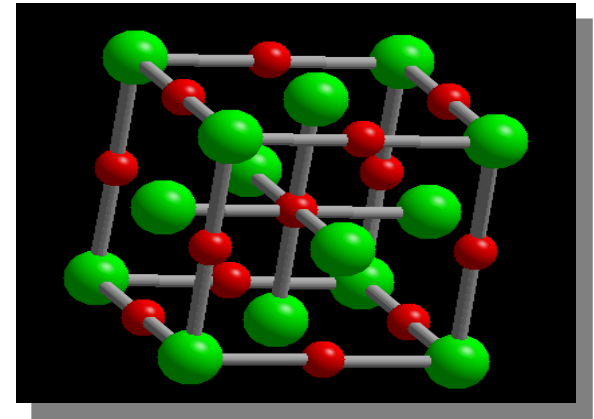
- Kinetic energy of the molecules arises from random vibrational, rotational and translational motion.



Potential Energy

- Potential energy: is the energy an object possesses by virtue of its position relative to other objects.
- Potential energy of molecules arises from intermolecular forces (Ionic, covalent and metallic bonds)

□ Potential energy can be converted into kinetic energy.



NaCl — composed of Na^+ and Cl^- ions.

- Electrostatic Potential energy (E_{el}): is the energy arises from the interactions between charged particles.

The diagram shows the equation $E_{el} = \frac{kQ_1Q_2}{d}$ inside a green rounded rectangle. Four arrows point from the equation to descriptive text: a left-pointing arrow from 'k' to 'constant of proportionality', a right-pointing arrow from 'Q₁Q₂' to 'electrical charges on the two interacting objects', a downward-pointing arrow from 'd' to 'distance', and a curved arrow from the top of the rectangle to the same text.

constant of proportionality

$E_{el} = \frac{kQ_1Q_2}{d}$

electrical charges on the two interacting objects

distance

- When Q_1 and Q_2 have the same sign → Repulsion → E_{el} positive.
- When Q_1 and Q_2 have opposite signs → Attraction → E_{el} negative.

Kinetic Energy and Potential Energy

- **Chemical energy:** is the potential energy stored in the arrangements of substances atoms which may be released during a chemical reaction.
- **Thermal energy:** the energy a substance possesses because of its temperature and is associated with the kinetic energy of the molecules in the substance.

UNITS OF ENERGY

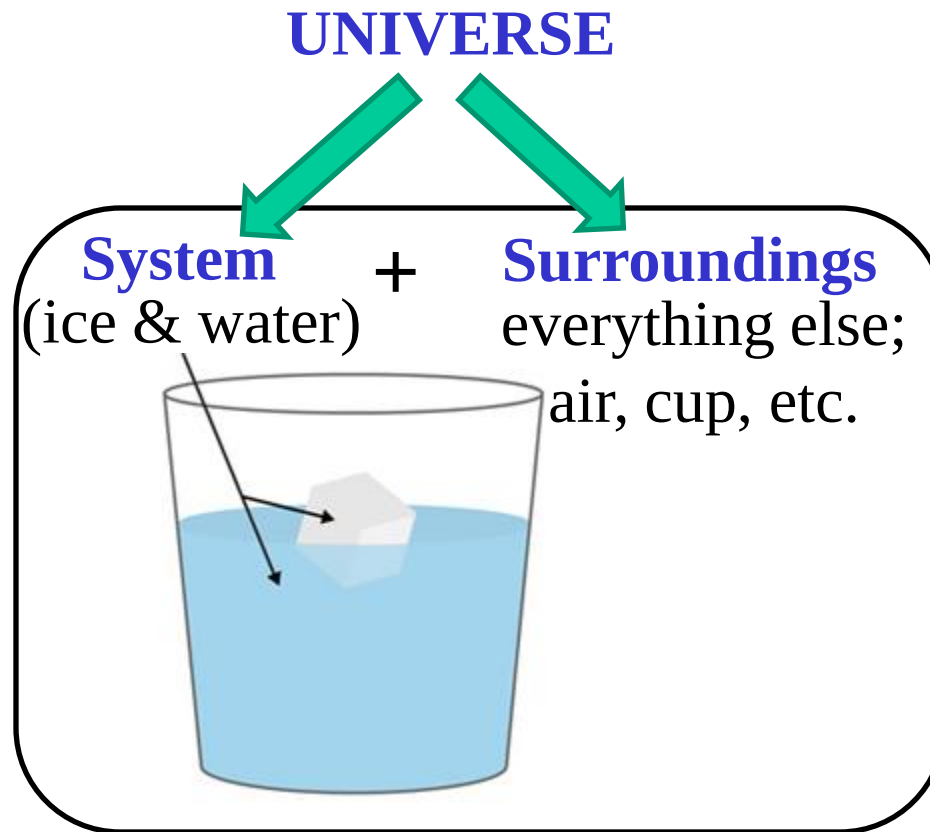
- SI Unit for energy is the joule, J.
- sometimes the calorie is **used** instead of the joule:

$$1 \text{ cal} = 4.184 \text{ J (exactly)}$$

Calorie: is the amount of energy required to raise the temperature of 1 g of water from 14.5°C to 15.5°C.

- A nutritional Calorie:
 $1 \text{ Cal} = 1000 \text{ cal} = 1 \text{ kcal}$

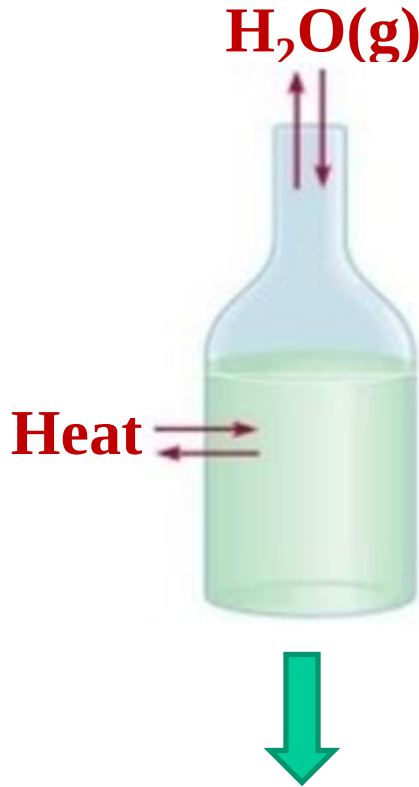
System and Surroundings



- **System:** part of the universe we are interested in.
- **Surroundings:** the rest of the universe.

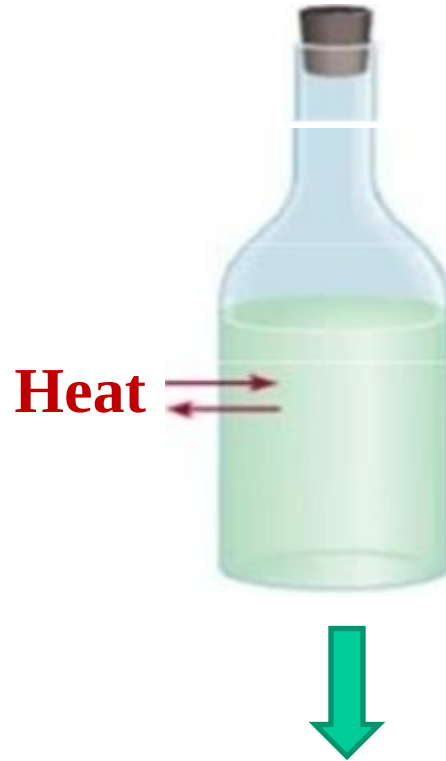
Systems

open system



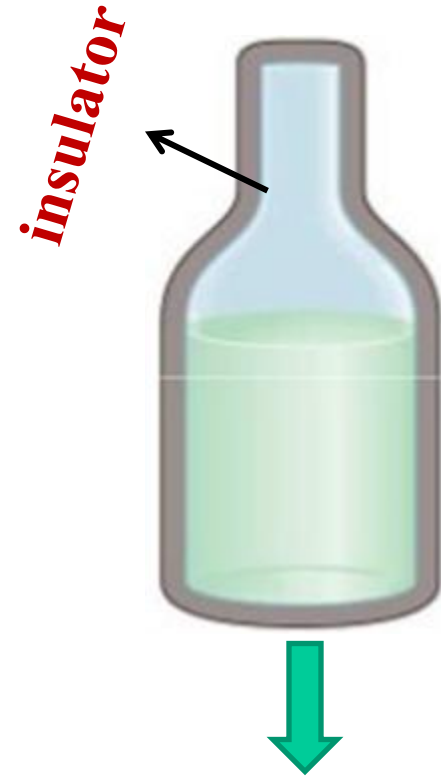
matter & energy
can be exchanged
with the
surroundings.

closed system



systems can
exchange energy
but not matter with
their surroundings.

isolated system



neither energy nor
matter can be
exchanged with the
surroundings.

Dr. Ziyad A. Taha

Transferring Energy

➤ System may exchange energy with its surroundings in two general ways:

1. **Work (w):** is defined as a **FORCE** causing the movement or displacement of an object.

➤ **Force (F):** is any push or pull exerted on an object

$$w = F.d \leftarrow \text{distance}$$

2. **Heat (q):** is the energy transferred from a hotter object to a colder one.

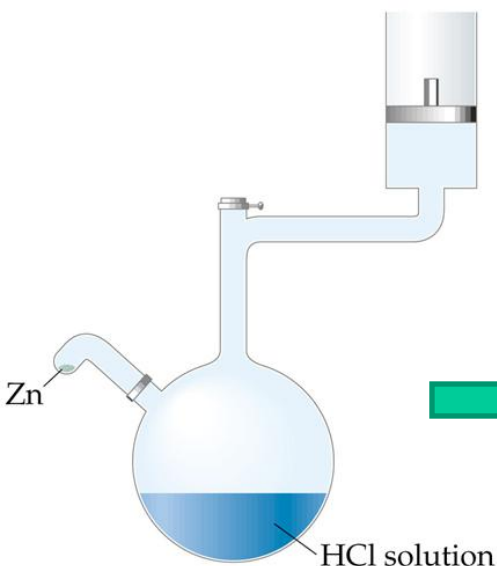
Force



**work done on
the system**



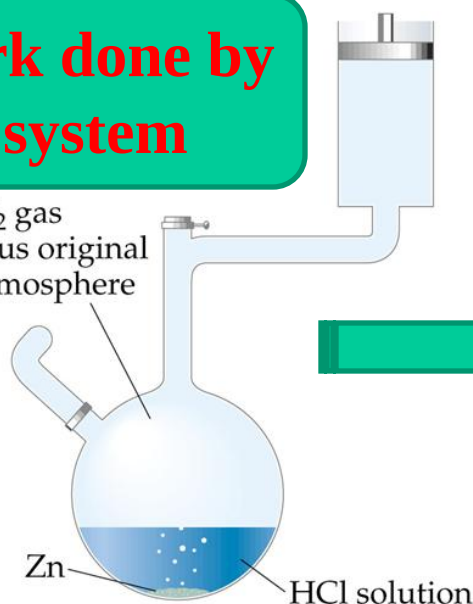
**Energy exchanges
with surr. as work**



**work done by
the system**

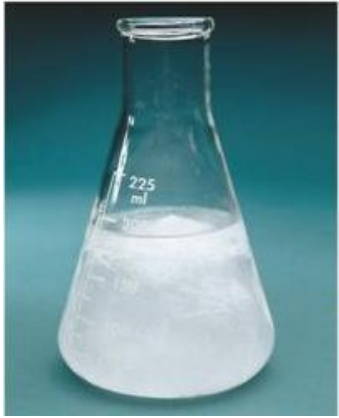


H₂ gas
plus original
atmosphere



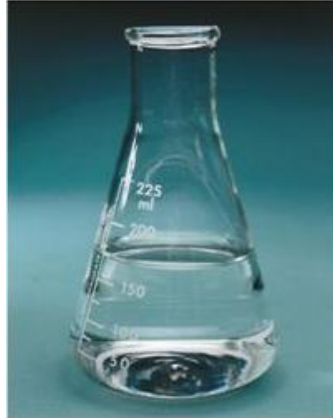
**Energy exchanges
with surr. as work**





Ice

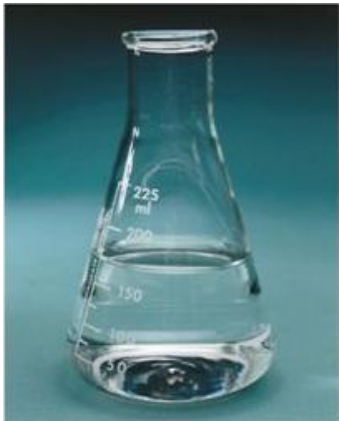
R.T
→



water

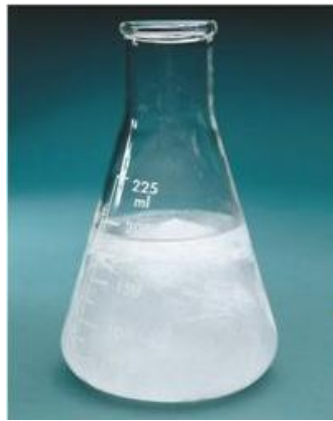


**Energy exchanges
with surr. as heat**



water

- 5°C
→



Ice



**Energy exchanges
with surr. as heat**

5.2 The First Law of Thermodynamics

- Any energy that is lost by the system must be gained by the surroundings, and vice versa.

$$\Delta E_{sys} = -\Delta E_{surr}$$

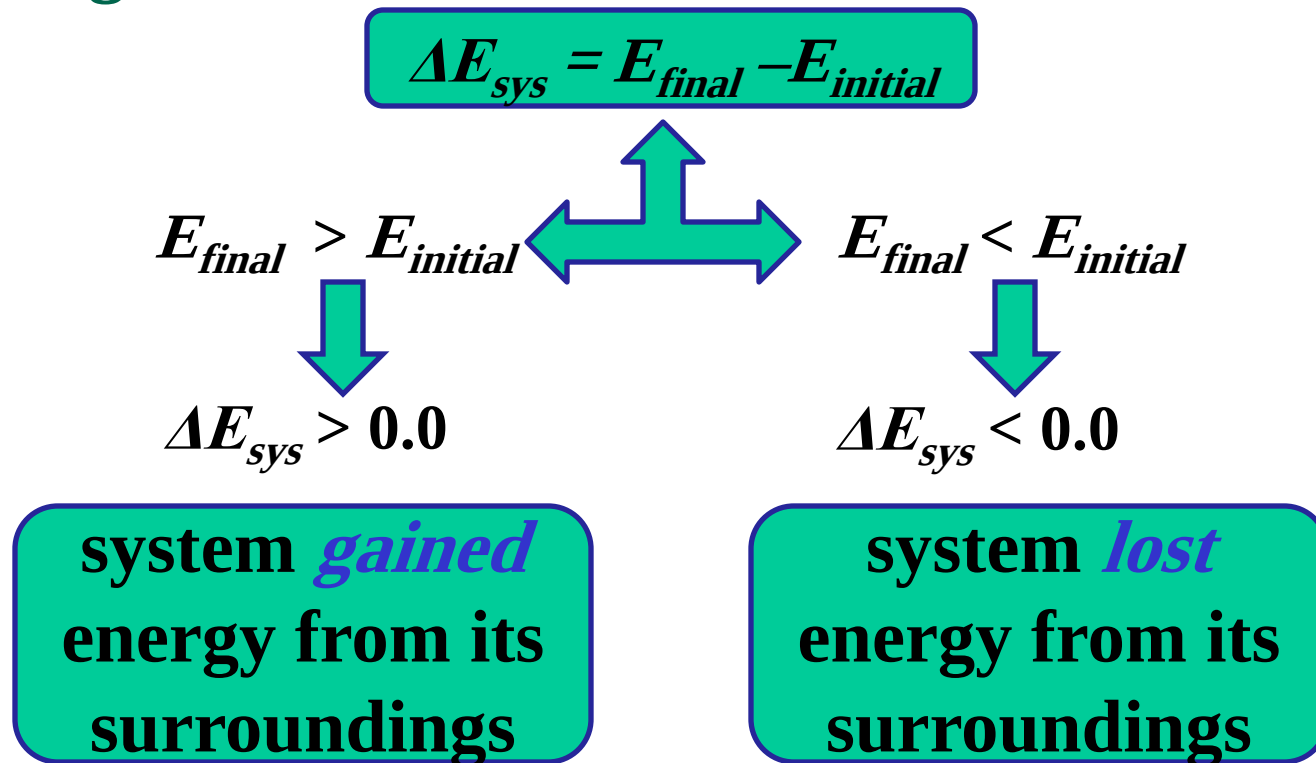
- Energy can be neither destroyed nor created.

Energy is conserved

$$\Delta E_{univ} = \Delta E_{sys} + \Delta E_{surr} = 0.0$$

Internal Energy

- **Internal Energy (E):** is the sum of all the kinetic and potential energies of the components of the system.
- Can not measure absolute internal energy.
 - In thermodynamics we are concerned with the change in E .



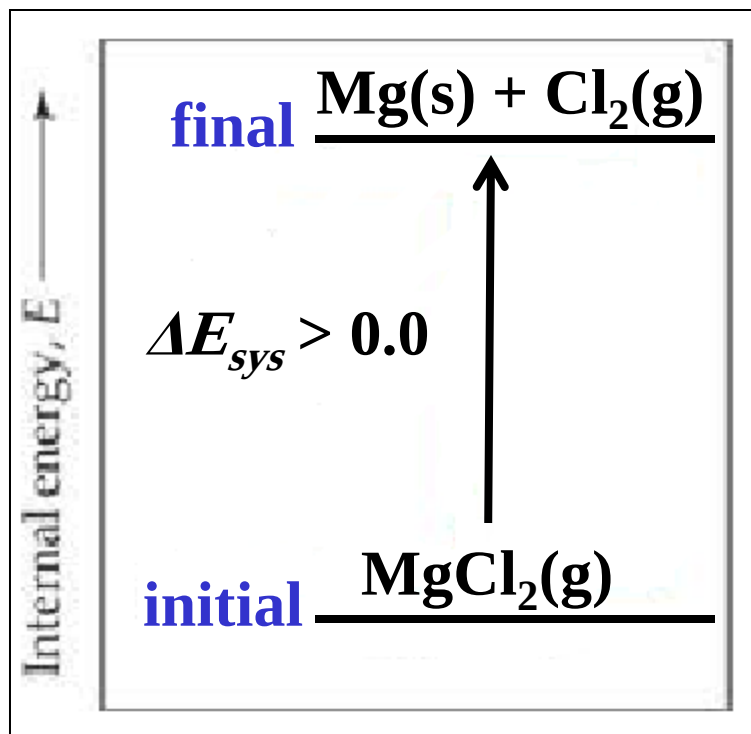
Ex: The internal energy for Mg(s) and Cl₂(g) is greater than that of MgCl₂(s). Sketch an energy diagram that represents the reaction



$E_{\text{products}} > E_{\text{reactants}}$

$E_{\text{final}} > E_{\text{initial}}$

$$\begin{aligned}\Delta E_{\text{sys}} &> 0.0 \\ \Delta E_{\text{surr}} &< 0.0 \\ \Delta E_{\text{univ}} &= 0.0\end{aligned}$$



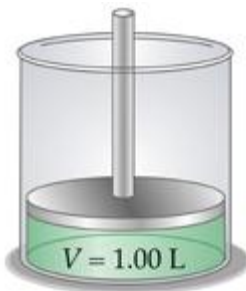
Relating ΔE to heat and work

- System may exchange energy with its surroundings in two general ways: w & q

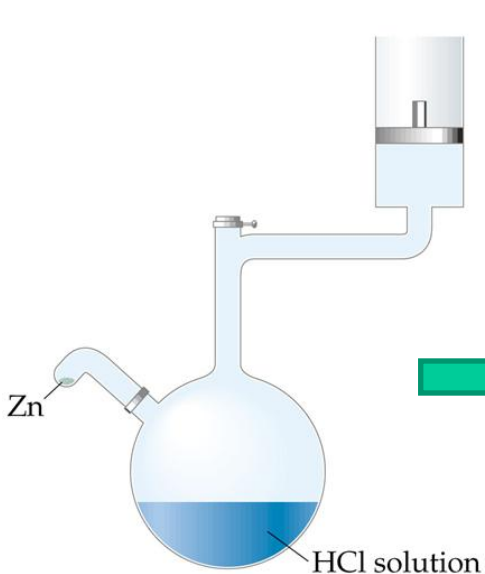
$$\Delta E_{sys} = q + w$$

- If $q(+)$ & $w(+)$ $\Rightarrow \Delta E_{sys} (+)$ net gain of energy by sys.
- If $q(-)$ & $w(-)$ $\Rightarrow \Delta E_{sys} (-)$ net loss of energy by sys.
- If $q(+)$ & $w(-)$ $\Rightarrow$ the sign of Δe_{sys} depends on the magnitude of q & w .
- If $q(-)$ & $w(+)$ $\Rightarrow$ the sign of Δe_{sys} depends on the magnitude of q & w .

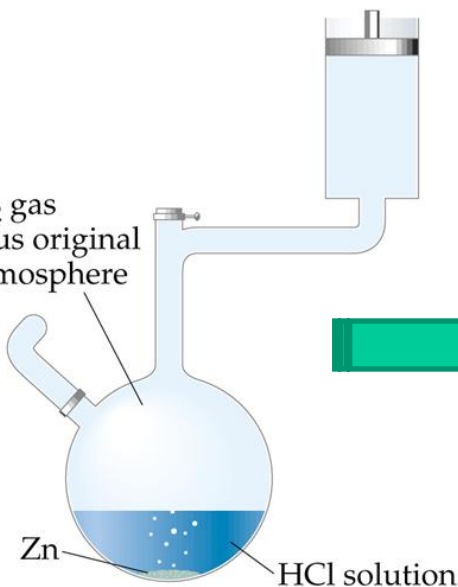
Force



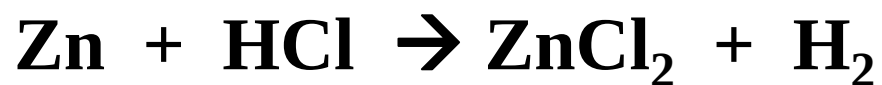
work done on the system $\rightarrow w(+)$



H₂ gas
plus original
atmosphere



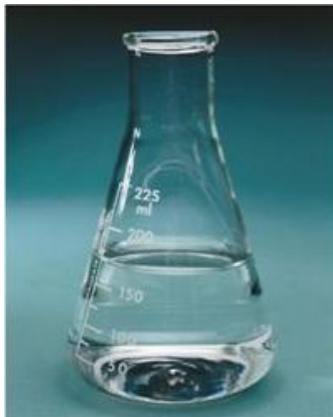
work done by the system $\rightarrow w(-)$





Ice

R.T
→



water



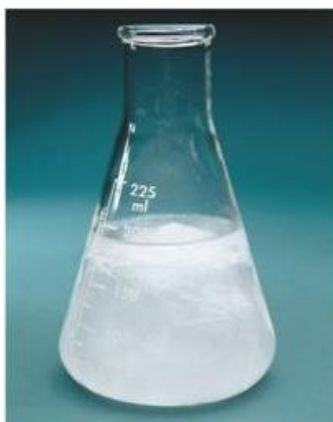
**System gains
heat → $q(+)$**

Endothermic



water

- 5°C
→



Ice

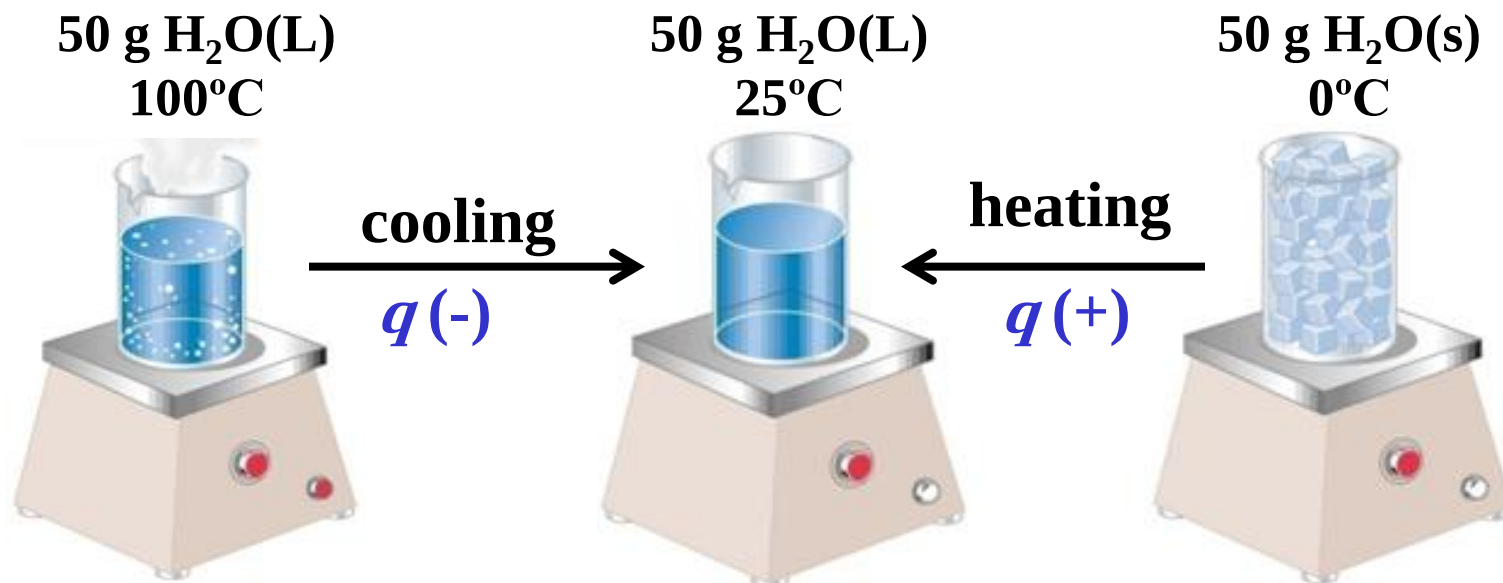


**System losses
heat → $q(-)$**

Exothermic

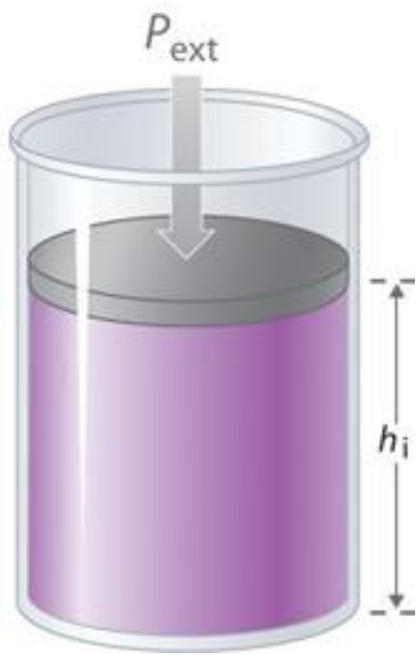
State Functions

- **State function:** its value depends only on the present state of the system, not on the path the system took to reach that state.



- E is a state function, so ΔE depends only on E_{final} & $E_{initial}$, not on how the change occurs.
- $E, P, V, T, \dots$ are state functions.
- q and w are not state functions.

Pressure-Volume Wo

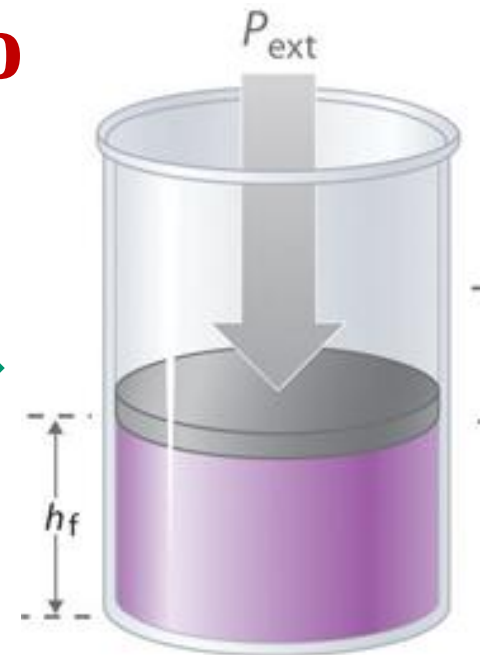


Work done on the
system $\{w(+)\}$



Compression

$w = ?$



$$w = F \bullet \Delta h$$

$$\Delta h = h_f - h_i \longrightarrow h_f < h_i \longrightarrow \Delta h_f \Rightarrow (-)$$

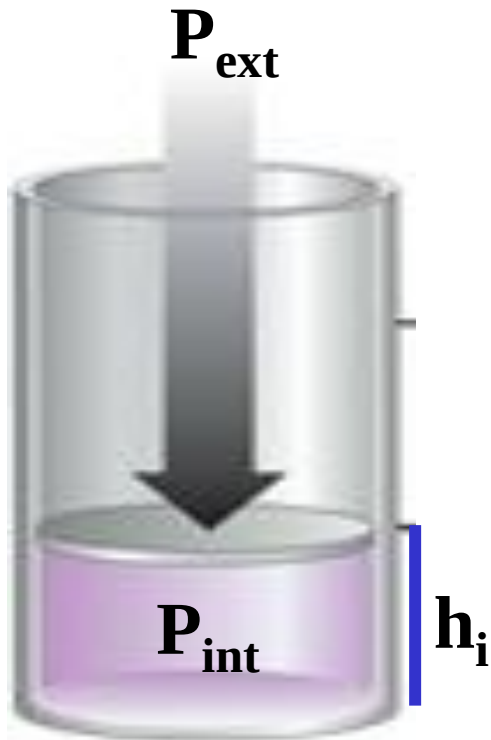
$$P = \frac{F}{A} \Rightarrow F = P \bullet A$$

P-V Work

$$w = P \bullet A \bullet (h_f - h_i)$$

$$w = P \bullet \Delta V \longrightarrow \Delta V \Rightarrow (-)$$

$$w = -P \bullet \Delta V$$



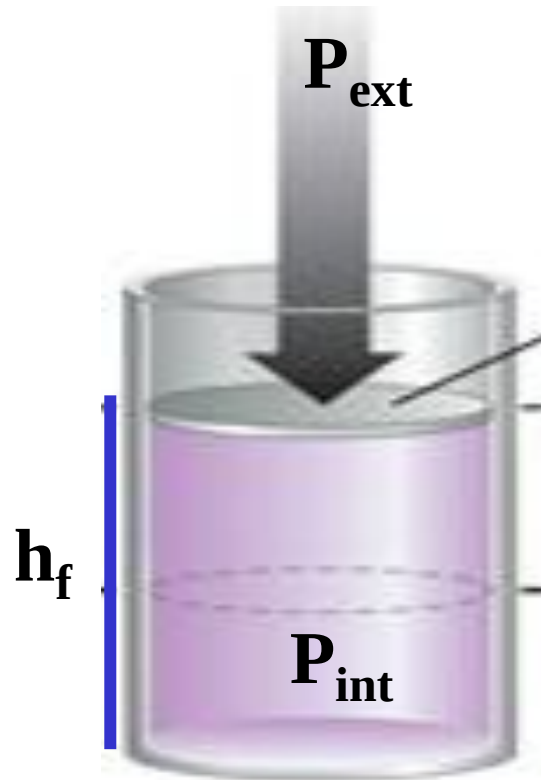
Work done by the
system $\{w(-)\}$

Expansion

$$w = ?$$

$$w = -P \bullet \Delta V$$

$$\Delta V \Rightarrow (+)$$



5.3 Enthalpy

- The chemical and physical changes that occurs around us occurs under constant atmospheric pressure.
- Chemical reactions can absorb or release heat and can be accompanied by work done by or on the system.

$$\Delta E_{\text{system}} = q + w$$

$$\Delta E = q_p - P\Delta V \quad \text{at constant } P$$

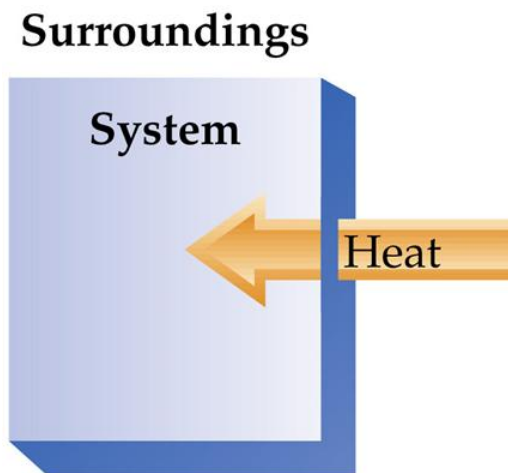
$$q_p = \Delta E + P\Delta V \longrightarrow q_p = \Delta H \quad \text{Enthalpy}$$

Enthalpy (H): Heat transferred between the system and surroundings carried out under constant pressure

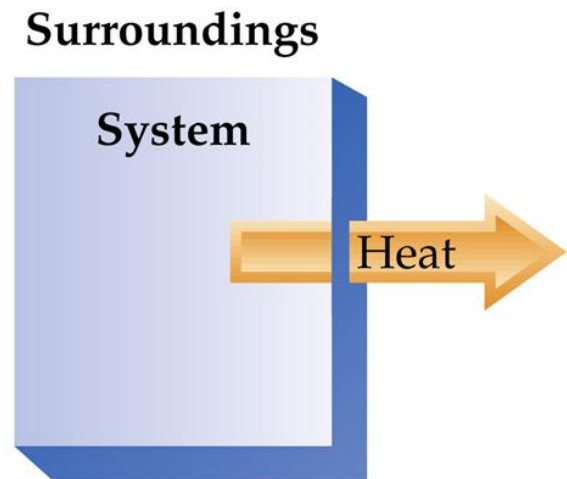
5.3 Enthalpy

$$\Delta H = \Delta E + P\Delta V$$

➤ Like {E, P, V}, H is a state function.



➤ The system has gained heat from the surroundings → ΔH is positive → Endothermic Process.



➤ The system has released heat to the surroundings → ΔH is negative → Exothermic Process

Sample Exercise:

Calculate the change in the internal energy and determine whether the process is endothermic or exothermic for the following cases:

- (a) a system releases 66.1 kJ of heat to its surroundings while the surroundings do 44.0 kJ of work on the system.**
- (a) a system absorbs 140 J of heat from the surroundings and does 85 J of work on the surroundings.**
- (a) a system absorbs 7.25 kJ of heat from the surroundings while its volume remains constant (assume that only P-V work can be done).**

5.4 Enthalpies of Reaction

- Since enthalpy is a state function:

$$\Delta H = H_{\text{final}} - H_{\text{initial}}$$

- For a chemical reaction:

$$\Delta H_{\text{rxn}} = H_{\text{products}} - H_{\text{reactants}}$$


The enthalpy of reaction
or
The heat of reaction

Exothermic reaction



$\Delta H = -483.6 \text{ kJ}$



Thermochemical Equation: balanced chemical equation associated with enthalpy change

5.4 Enthalpies of Reaction

- Enthalpy is an *extensive* property (magnitude of ΔH is directly proportional to amount):



- $\Delta H_{\text{rxn}} = -\Delta H_{\text{reverserxn}}$



5.4 Enthalpies of Reaction

➤ ΔH_{rxn} depends on the state of the reactants and products.



Sample Exercise:

Consider the following reaction:



(a) Calculate the amount of heat transferred when 3.55 g of Mg reacts at constant pressure

{-87.9 kJ}

(b) How many grams of MgO are produced during an enthalpy change of -234 kJ?

{15.7 g}

(c) How many kJ of heat are absorbed when 40.3 g of MgO is decomposed into Mg and O₂ at constant pressure?

{602 kJ}

Sample Exercise:

Consider the following reaction:



(a) Calculate ΔH for:



(b) Calculate ΔH for:



5.5 Calorimetry

- The value of ΔH can be determined experimentally by measuring the heat flow accompanying a reaction at constant P.
- **Calorimetry** = measurement of heat flow.
- **Calorimeter** = a device used to measure heat flow.
- Heat capacity(C) = the amount of heat required to raise the temperature of a substance by 1 K or 1°C.

$$C = \frac{q}{\Delta T}$$

J/K
or
 $J/^\circ C$

heat, J

temperature change
K or $^\circ C$

$$q = C \cdot \Delta T$$

- Specific heat (C_s) = the amount of heat required to raise the temperature of one gram of a substance by one degree (1 K or 1°C).

$$C_s = \frac{q}{m \cdot \Delta T}$$

Diagram illustrating the components of the specific heat equation:

- C_s is labeled with units: J/g. K or J/g.°C.
- q is labeled: heat, J.
- $m \cdot \Delta T$ is labeled: temperature change K or °C.
- m is labeled: Mass, g.

$$q = C_s \cdot m \cdot \Delta T$$

$$q = C \cdot \Delta T$$

- Molar heat capacity (C_m) = the amount of heat required to raise the temperature of one mole of a substance by 1 K or 1°C.

$$C_m = \frac{q}{n \cdot \Delta T}$$

J/mol. K
 or
 J/mol.°C

heat, J
 temperature change
 K or °C

mol

$$q = C_m \cdot n \cdot \Delta T$$

$$C_m = C_s \cdot \text{Molar mass}$$

Sample Exercise:

If 209 J is required to increase the temperature of 50 g of water by 1.0 K, calculate the specific heat of water.

$$C_s = \frac{q}{m \times \Delta T} \longrightarrow C_s = \frac{209}{50 \times 1.0}$$

$$C_s = 4.18 \text{ J/g.}^\circ\text{C} = 4.18 \text{ J/g.K}$$

Sample Exercise:

If 24.2 kJ is used to warm a piece of aluminum with a mass of 250. g, what is the final temperature of the aluminum if its initial temperature is 5.0°C?

$$C_s = \frac{q}{m \times \Delta T} \longrightarrow 0.90 = \frac{24200}{250 \times \Delta T}$$

$$\Delta T = 107^\circ\text{C} \longrightarrow T_f = 107 - 5 = 102^\circ\text{C}$$

❖ For same sample of water, $T_f = 23^\circ\text{C}$.

Specific Heats (S) of Some Substances at 298 K

Substance	C_s (J g ⁻¹ K ⁻¹)	Substance	C_s (J g ⁻¹ K ⁻¹)
N ₂ (g)	1.04	H ₂ O(l)	4.184
Al(s)	0.902	H ₂ O(s)	2.06
Fe(s)	0.45	CH ₄ (g)	2.20
Hg(l)	0.14	CO ₂ (g)	0.84

- Which substance in above table requires the smallest amount of energy to increase the temperature of 50 g of that substance by 10 K?
- What is the molar heat capacity of CO₂(g)?
{36.96 J/mol}
- How many kJ of heat are needed to raise the temperature of 10.0 Kg of Al(s) from 24. 6 K to 46.2 K?{194.8 kJ}

Calorimetry

➤ Main idea of calorimetry is that:

heat lost by system = - heat gained by surroundings

$$q_{\text{sys}} = -q_{\text{surr}}$$

$$(C_s \bullet m \bullet \Delta T)_{\text{sys}} = -(C_s \bullet m \bullet \Delta T)_{\text{surr}}$$

Sample Exercise:

If a 30.0 g piece of copper pipe at 80.0°C is placed in 100.0 g of water at 27.0°C, what is the final temperature? *{Assume that no heat is transferred to the surroundings. C_s for water = 4.18 J/g.K and for copper = 0.385 J/g.K }*

$$T_f = 28.4\text{ }^{\circ}\text{C}$$

Sample Exercise:

A 28.0 g chunk of aluminum is dropped into 100.0 g of water with an initial temperature of 20.0 °C. If the final temperature of the water is 24.0 °C, what was the initial temperature of the aluminum? *{Assume that no heat is transferred to the surroundings. C_s for water = 4.18 J/g.K and for copper = 0.902 J/g.K }*

$$T_f = 90.6^\circ\text{C}$$

- **Coffee-cup calorimeter:** measures the heat flow for reactions that occur at constant pressure ($q_p = \Delta H$).
 - Acid-Base reactions.
 - Precipitation reactions.
 - Oxidation reduction reactions.

Coffee-cup calorimeter

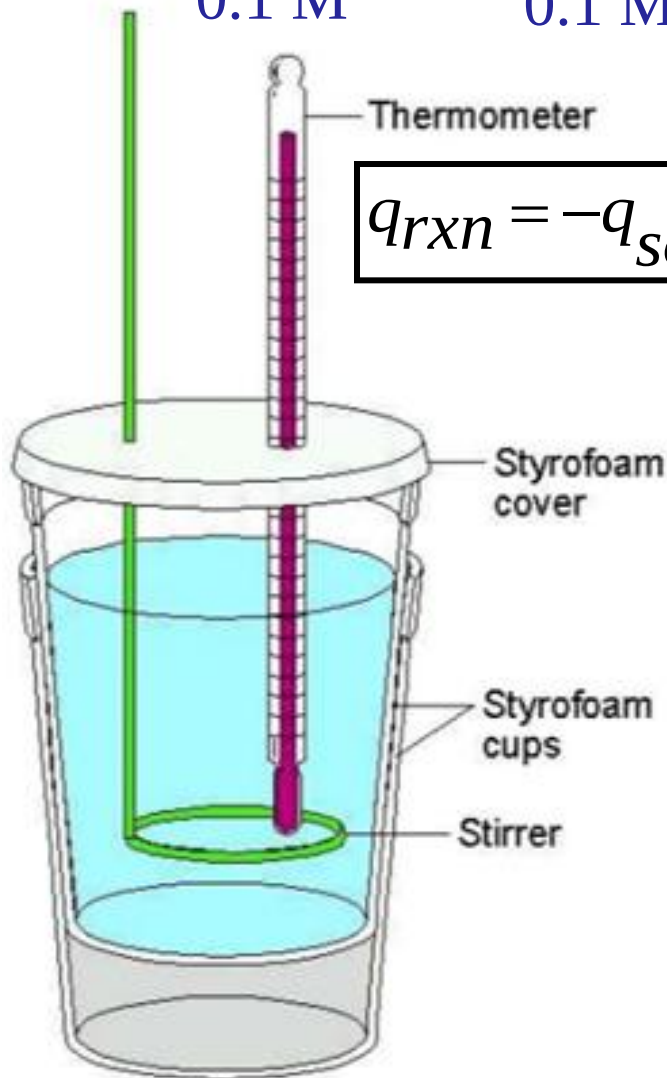


50 ml
0.1 M

50 ml
0.1 M

Thermometer

$$q_{\text{rxn}} = -q_{\text{solution}} = -C_s \cdot (\text{total mass of solution}) \cdot \Delta T$$

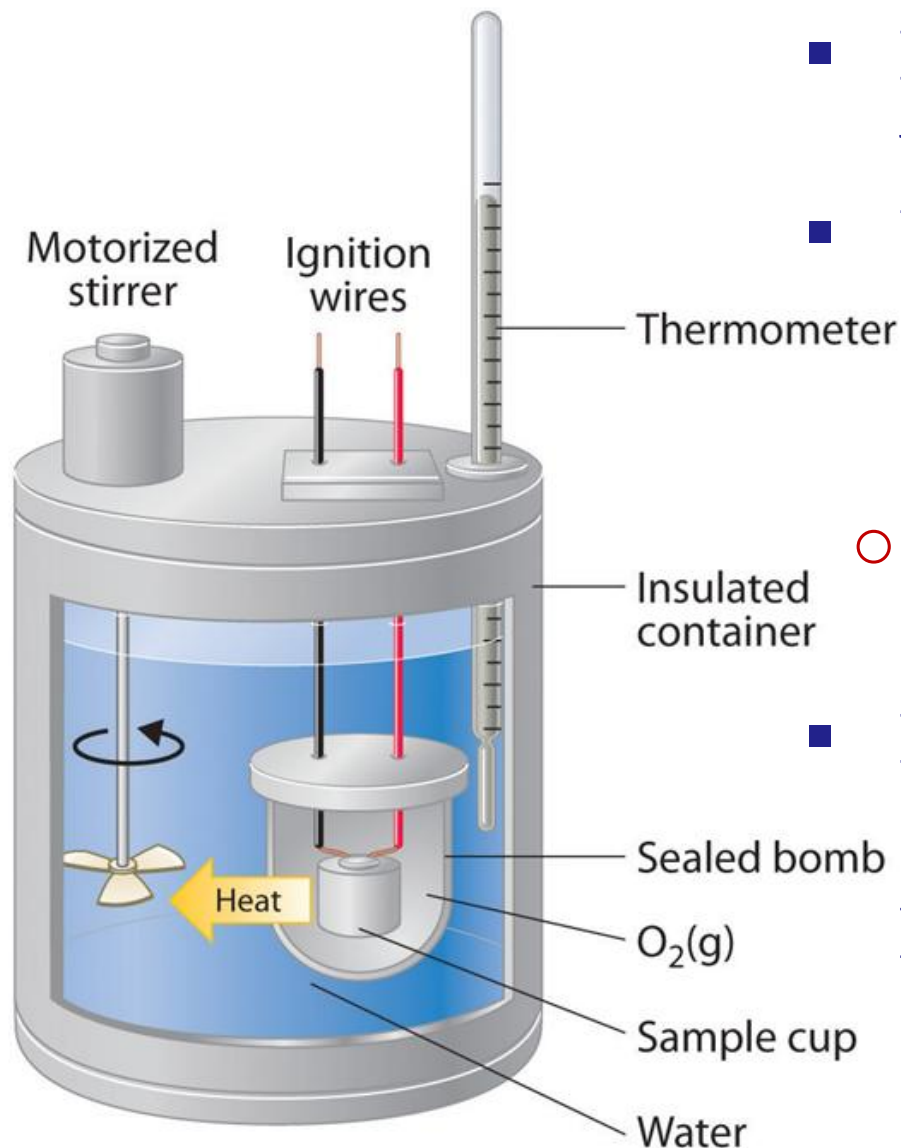


Total mass = total volume x $d_{\text{H}_2\text{O}}$

Total mass = 100 mL x 1.0 g/mL

$C_s = 4.18 \text{ J/mol.K}$

Bomb Calorimetry



- Heat flow at constant volume $\{q_v = \Delta E \cong \Delta H\}$.
- For combustion reactions.

$$q_{\text{rxn}} = -C_{\text{cal}} \times \Delta T$$

○ C_{cal} : total heat capacity of the calorimeter

- Measurements made with a bomb are generally more precise than those made by coffee-cup.

Sample Exercise:

50.0 mL of 1.0 M HCl at 25.0°C were mixed with 50.0 mL of 1.0 M NaOH also at 25.0°C in a styrofoam cup calorimeter. After the mixing process, the thermometer reading was at 31.9°C. *[assuming that the calorimeter loses only a negligible quantity of heat, that is the density is 1.0 g/mL, and that its specific heat is 4.18 J/g.K]*
Calculate:

(a) the energy involved in the reaction.

{-2.9 kJ}

(b) the enthalpy per moles of hydrogen ions used.

{-58 kJ/mol}

[heat of neutralization for strong acid/base reactions]

Sample Exercise:

In an experiment, 1.50 g of Mg(s) was combined with 125.0 mL of 1.0 M HCl. The initial temperature was 25.0°C and the final temperature was 72.3°C. Calculate:

(a) the heat involved in the reaction.

{−25.0 kJ }

(b) the enthalpy of reaction in terms of the number of moles of Mg(s) used.

{−406 kJ/mol}

Sample Exercise:

When 5.03 g of solid potassium hydroxide are dissolved in 100.0 mL of distilled water in a coffee-cup calorimeter, the temperature of the liquid increases from 23.0°C to 34.7°C. The density of water in this temperature range averages 0.9969 g/cm³. What is ΔH_{soln} (in kJ/mole)? *{Assume that the calorimeter absorbs a negligible amount of heat and, because of the large volume of water, the specific heat of the solution is the same as the specific heat of pure water $C_s = 4.18 \text{ J/g.K .}$ }*

-57.2 KJ/mol

Sample Exercise:

The combustion of 0.579 g of benzoic acid in a bomb calorimeter caused a 2.08 °C increase in the temperature of the calorimeter. The chamber was then emptied and recharged with 1.732 g of glucose and excess oxygen. Ignition of the glucose resulted in a temperature increase of 3.64°C. What is the ΔH_{comb} of glucose? {Assume that no heat is transferred to the surroundings, ΔH_{comb} of benzoic acid = -26.38kJ/g }

$$2.78 \times 10^3 \text{ J/mol}$$

Hess's Law

- Hess's law: if a reaction is carried out in a series of steps, ΔH for the overall reaction equal the sum of ΔH for the individual step.

For example:



$$\Delta H = \Delta H_1 + \Delta H_2$$

$$\Delta H = -802 + -88 = -890 \text{ kJ}$$

Sample Exercise:

What is the value for the heat of combustion, ΔH_c , of the following reaction?



Given:



Sample Exercise:

reverse equation 3



$$-\Delta H_3 = -87.9 \text{ kJ}$$

multiply equation 2 by 2



$$2\Delta H_2 = -593.6 \text{ kJ}$$

equation 1 as is



$$\Delta H_1 = -393.5 \text{ kJ}$$



$$\Delta H = \Delta H_1 + 2\Delta H_2 + -\Delta H_3$$

$$\Delta H = -393.5 + -593.6 - 87.9$$

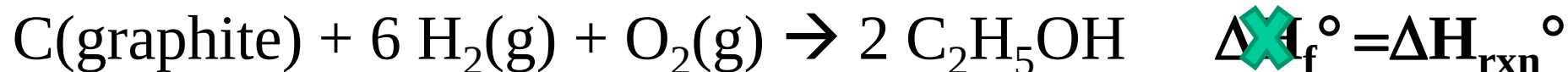
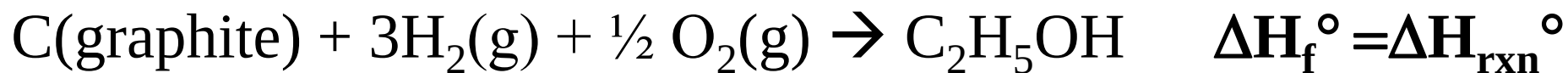
$$\Delta H = -1075 \text{ kJ}$$

5.7 Enthalpies of Formation

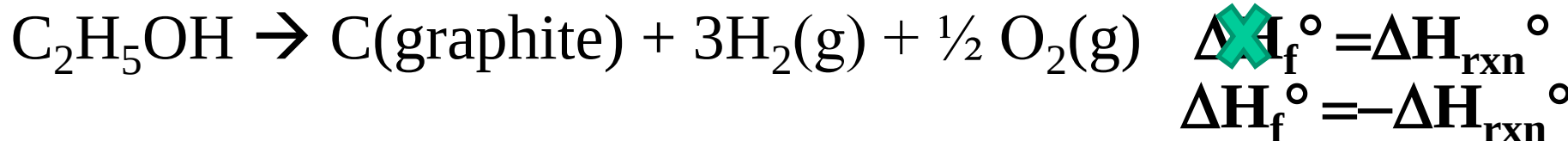
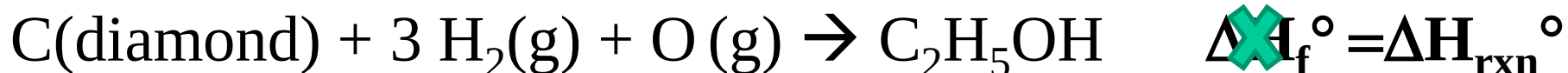
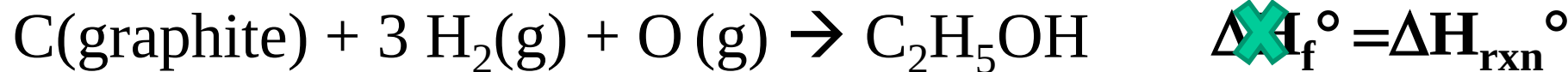
- ΔH_f **Enthalpies of formation or heat of formation is defined as:** Enthalpy change associated with the formation of a compound from its constituent elements.
- $\Delta H_{\text{rxn}}^\circ$ **Standard enthalpy change of reaction is defined as:** the enthalpy change when all reactants and products are in their standard states.
- **Standard conditions (standard state):** Most stable form of the substance (its pure state) at 1 atm and the temperature of interest usually 25°C (298 K).

5.7 Enthalpies of Formation

- ΔH_f° Standard enthalpy of formation of a compound is defined as the change in enthalpy for the reaction that forms one mole of the compound from its elements in their most stable form at standard states.



$$2\Delta H_f^\circ = \Delta H_{\text{rxn}}^\circ$$



5.7 Enthalpies of Formation

- ΔH_f° The standard enthalpy of formation of the most stable form of any element is zero.



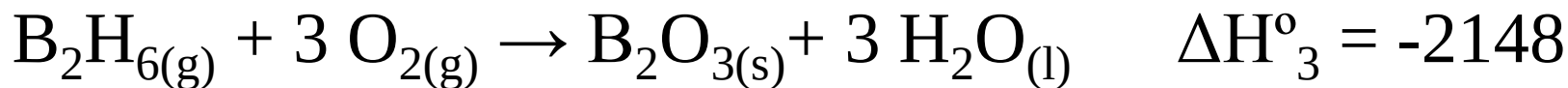
- For a reaction

$$\Delta H_{\text{rxn}}^\circ = \sum n \cdot \Delta H_f^\circ (\text{products}) - \sum m \cdot \Delta H_f^\circ (\text{reactants})$$

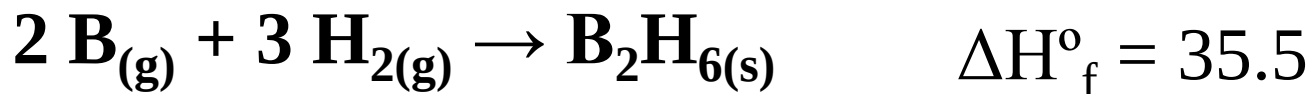
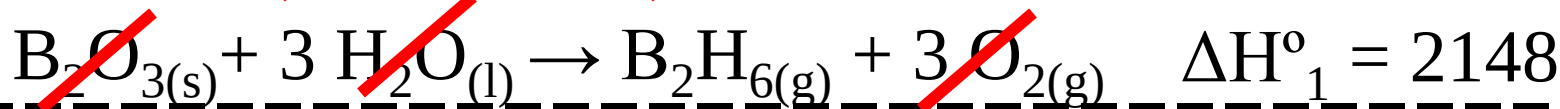
- note: n & m are stoichiometric coefficients.

Sample Exercise:

Calculate ΔH° for the formation of $\text{B}_2\text{H}_6(\text{g})$, given that:



1. Reverse equation 3.
2. Multiply equation one by 1/2.
3. Multiply eq 2 by 3/2.



Periodic Table of the Elements

IA																	VIIIA									
1 H 1.0079																	2 He 4.0026									
3 Li 6.941	4 Be 9.0122																	5 B 10.81	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.179			
11 Na 22.990	12 Mg 24.305																	13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.06	17 Cl 35.453	18 Ar 39.948			
19 K 39.098	20 Ca 40.08	21 Sc 44.956	22 Ti 47.90	23 V 50.941	24 Cr 51.996	25 Mn 54.938	26 Fe 55.847	27 Co 58.933	28 Ni 58.71	29 Cu 63.546	30 Zn 65.38	31 Ga 69.72	32 Ge 72.59	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80									
37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.22	41 Nb 92.906	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.4	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.90	54 Xe 131.30									
55 Cs 132.91	56 Ba 137.33	71 Lu 174.97	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.21	76 Os 190.2	77 Ir 192.22	78 Pt 195.09	79 Au 196.97	80 Hg 200.59	81 Tl 204.37	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)									
87 Fr (223)	88 Ra 226.03	103 Lr (260)	104 (261)	105 (262)	106 (263)	*Name Not Officially Assigned																				

Alkali Metals

Alkaline Earth Metals

Transition Metals

Other Metals

Nonmetals

Noble Gases

Inner Transition Metals

Gaseous State

Liquid State

Solid State

Synthetically Prepared

Lanthanide Series	57 La 138.91	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.4	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04
Actinide Series	89 Ac (227)	90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (254)	100 Fm (257)	101 Md (258)	102 No (259)

Sample exercise

Calculate heat of reaction for the combustion of propane gas giving carbon dioxide and water.



$$\Delta H^\circ_{\text{f}}: \quad -103.85 \quad \quad \quad -393.5 \quad \quad \quad -285.8$$

$$\Delta H^\circ_{\text{rxn}} = \{4\Delta H^\circ_{\text{f}}[\text{H}_2\text{O}] + 3\Delta H^\circ_{\text{f}}[\text{CO}_2]\} - \{5\Delta H^\circ_{\text{f}}[\text{O}_2] + \Delta H^\circ_{\text{f}}[\text{C}_3\text{H}_8]\}$$

$$\Delta H^\circ_{\text{rxn}} = \{4 \cdot -285.8 + 3 \cdot -393.5\} - \{5 \cdot 0.0 + 1 \cdot -103.85\}$$

$$\Delta H^\circ_{\text{rxn}} = -2220 \text{ kJ}$$

Sample exercise

Complete combustion of one mole of $\text{C}_3\text{H}_6\text{O}$ liberates 1790 kJ. Given that the $\Delta H^\circ_{\text{f}}[\text{CO}_2] = -393.5$ and $\Delta H^\circ_{\text{f}}[\text{H}_2\text{O}] = -285.8$ kJ/mol.



$$\Delta H^\circ_f: \quad ??? \qquad \qquad -393.5 \qquad \qquad -285.8$$

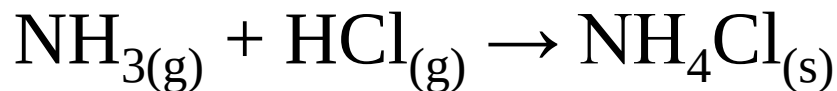
$$\Delta H^{\circ}_{\text{rxn}} = \{3\Delta H^{\circ}_{\text{f} [\text{H}_2\text{O}]} + 3\Delta H^{\circ}_{\text{f} [\text{CO}_2]}\} - \{4\Delta H^{\circ}_{\text{f} [\text{O}_2]} + \Delta H^{\circ}_{\text{f} [\text{C}_3\text{H}_6\text{O}]}\}$$

$$-1790 = \{3 \cdot -285.8 + 3 \cdot -393.5\} - \{5 \cdot 0.0 + \Delta H^\circ_{f [C_3H_6O]}\}$$

$$\Delta H^{\circ}_{f [C_3H_6O]} = -248.9 \text{ kJ/mol}$$

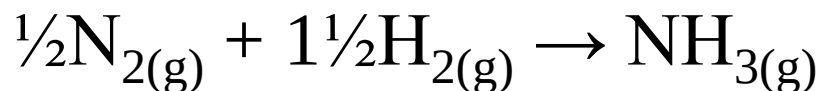
Sample Exercise:

Calculate ΔH for the reaction:

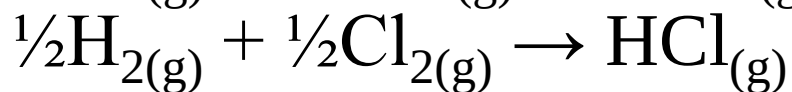


$\Delta H = ??$

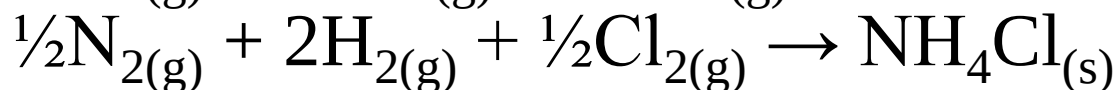
Given that:



$$\Delta H = -46.1 \text{ kJ mol}^{-1}$$



$$\Delta H = -92.3 \text{ kJ mol}^{-1}$$



$$\Delta H = -314.4 \text{ kJ mol}^{-1}$$

Thermochemistry

$$E_k = \frac{1}{2} \cdot m \cdot v^2 \quad (kg \, m^2 \, s^{-2} \equiv \text{joule})$$

$$\Delta E = q + w$$

Energy

1st Law of Thermodynamics

$$\Delta H = q_P$$

$$q_{\text{rxn}} = -q_{\text{soln}} = -(\text{specific heat of solution}) \times (\text{grams of solution}) \times \Delta T$$

Enthalpy / Calorimetry

Hess' Law

Enthalpy of Formation

$$\Delta H^\circ_{\text{rxn}} = \sum n \cdot \Delta H^\circ_f (\text{products}) - \sum m \cdot \Delta H^\circ_f (\text{reactants})$$